

# Firmware, BIOS, UEFI & Drivers

## Lesson 46 — The Boot Firmware Layer

JKSSB Computer Awareness

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# What Is Firmware?

- ▶ **Firmware** is a special kind of software stored in **non-volatile** memory on the hardware itself.
- ▶ It gives a device its basic, low-level startup instructions.
- ▶ It runs **before** the operating system and starts the hardware.
- ▶ Firmware is programmed into a read-only memory (ROM) or a flash chip, not loaded from a file each time.

## The one idea

Firmware sits between hardware and higher-level software; without it the hardware does not know how to start.

# Where Firmware Is Stored

- ▶ Firmware is kept in **non-volatile** memory, so it survives when the power is off.
- ▶ Common locations: **ROM** (Read-Only Memory), PROM, EPROM, EEPROM, and **flash** memory.
- ▶ Because it is non-volatile, the startup instructions are ready the moment the machine is switched on.
- ▶ This is why a computer can begin working before any file is read from disk.

## Trap alert

Firmware lives in ROM/flash, **not** in RAM. RAM is volatile and is empty at power-on.

# System Firmware: The BIOS

- ▶ **BIOS** stands for **Basic Input/Output System**.
- ▶ It is the firmware stored on the motherboard of a PC.
- ▶ It runs **first** when the computer is switched on.
- ▶ It **initialises the hardware** and then **loads the operating system**.

## Exam fact

The firmware that initialises hardware and loads the OS is the BIOS (PYQ-28).

# The Power-On Boot Sequence

► The machine follows a fixed chain each time it starts:

1. Power on.
2. Firmware (**BIOS/UEFI**) runs.
3. Hardware initialisation and **POST**.
4. Firmware finds and loads the **operating system**.
5. The OS starts and loads **device drivers**.
6. **Application** programs run.

power-on → firmware → initialisation → OS → driver → application

# POST: Power-On Self-Test

- ▶ **POST** stands for **Power-On Self-Test**.
- ▶ It is one of the first things the BIOS does after power-on.
- ▶ It checks that key hardware (CPU, RAM, keyboard, disks) is present and working.
- ▶ If POST fails, the computer may beep or show an error instead of booting.

## Trap alert

POST is a BIOS step and runs **before** the operating system, not after it.

# What the BIOS Does

- ▶ Runs the **POST** to test the basic hardware.
- ▶ **Initialises** and configures the basic hardware.
- ▶ Reads settings such as the **boot order**.
- ▶ Finds the boot device and hands control to the **operating system**.
- ▶ Provides basic input/output services to the OS at an early stage.

# BIOS Setup and CMOS

- ▶ BIOS settings — date/time, boot order, passwords — are stored in a small battery-backed chip called **CMOS**.
- ▶ The BIOS Setup utility is opened with a key during startup, commonly **Del** or **F2**.
- ▶ Keeping these settings on a battery-backed chip is why the clock and boot order are remembered between sessions.

The exact setup key varies by manufacturer.

# UEFI: The Modern Firmware

- ▶ **UEFI** stands for **Unified Extensible Firmware Interface**.
- ▶ It is the modern replacement for the legacy BIOS.
- ▶ It starts the machine **faster**, supports large disks (GPT), and offers a graphical setup utility.
- ▶ It adds **Secure Boot**, which helps block unauthorised software from loading at startup.

## For the exam

The firmware that boots the machine may be called BIOS or UEFI depending on the age of the system.

# BIOS vs UEFI

| Point        | BIOS                      | UEFI                                  |
|--------------|---------------------------|---------------------------------------|
| Full form    | Basic Input/Output System | Unified Extensible Firmware Interface |
| Age          | Legacy (older)            | Modern                                |
| Boot speed   | Slower                    | Faster                                |
| Disk support | MBR (about 2 TB limit)    | GPT (very large disks)                |
| Interface    | Mostly text-based         | Graphical                             |
| Security     | No Secure Boot            | Secure Boot available                 |

# Device Drivers

- ▶ A **device driver** is **software** that lets the operating system talk to a specific hardware device.
- ▶ Each device — printer, graphics card, network card — needs its own driver.
- ▶ The driver translates general OS commands into device-specific instructions.
- ▶ Drivers are loaded **by the operating system**, not before it.

## Keep it straight

A driver is software, **not** hardware, and it works **after** the OS has started.

# Drivers in the Boot Chain

- ▶ Sequence: firmware boots, the **OS** starts, then the OS loads the **drivers**.
- ▶ **BIOS/UEFI** starts the machine; drivers run only once the OS is running.
- ▶ A driver is matched to the exact device and OS version.

## Trap alert

The layer that first initialises hardware and loads the OS is **BIOS/UEFI**, never the device driver (TRAP-24).

# Firmware vs Driver vs Operating System

| Layer            | Stored                      | Runs                | Role                                     |
|------------------|-----------------------------|---------------------|------------------------------------------|
| Firmware         | On the hardware (ROM/flash) | First, at power-on  | Starts the machine; loads the OS         |
| Device driver    | On disk, with the OS        | After the OS starts | Talks to one device                      |
| Operating system | On disk                     | After firmware      | Manages resources and the user interface |

Keep the three roles separate — a favourite exam confusion.

# Firmware Updates (Flashing)

- ▶ A **firmware update** replaces the firmware stored on a device.
- ▶ The process is commonly called **flashing**.
- ▶ Updates fix bugs, add features, and patch security flaws.
- ▶ Firmware updates are much **rarer** than ordinary software updates.

## Best practice

Always use the manufacturer's update for the exact model and keep a stable power supply during the update.

# Risks of a Firmware Update

- ▶ Firmware is low-level, so a failed update can leave a device unusable.
- ▶ Interrupting power during flashing can “brick” the device.
- ▶ Loading the **wrong firmware** for the model can damage it.

## Warning

Never switch off or restart the machine in the middle of a firmware update.

## Worked Example

### A PC reports “no boot device found”

Follow the boot chain. The firmware (BIOS/UEFI) runs, then the POST passes, then the firmware looks for a boot device. It cannot find the operating system, so it stops and shows the message. The failing layer is the **firmware/boot configuration**, not the application.

Walk the chain in order: firmware → POST → OS → driver → application.

## Trap Alert: Keep the Boot Roles Straight

- ▶ Firmware that initialises hardware and loads the OS is **BIOS/UEFI**, not a driver or kernel (TRAP-24, PYQ-28).
- ▶ A **device driver** is software; the OS is system software (TRAP-10).
- ▶ Firmware is stored in ROM/flash (non-volatile); RAM is volatile.
- ▶ **POST** runs before the OS, not after.
- ▶ A driver is loaded **by the OS**, after the firmware.

## Lesson 46: Recall

- ▶ **Firmware** is software in non-volatile memory (ROM/flash) that starts and controls hardware.
- ▶ **BIOS** = Basic Input/Output System; it initialises hardware and loads the OS.
- ▶ **POST** = Power-On Self-Test, run by the BIOS before the OS.
- ▶ **UEFI** = Unified Extensible Firmware Interface; the modern firmware with fast boot, GPT, and Secure Boot.
- ▶ The startup chain is power-on → firmware → initialisation → OS → driver → application.
- ▶ A **device driver** is software loaded by the OS; firmware is not a driver.
- ▶ A firmware update is called **flashing**; interrupting it can damage the device.